

Real Estate Voice Agent

An n8n backend that powers a Retell phone voice agent for a real estate office — searching listings, booking viewings, taking new listings, and escalating sensitive calls to a human.

n8n

Retell

Google Sheets

Gmail

1 Executive Summary

A real estate office fields constant phone traffic: buyers asking what's available, owners wanting to list a property, people booking viewings, and callers who just want a human. This backend gives a Retell voice agent five callable tools that handle exactly these requests, each backed by validation logic so the agent searches real inventory, never double-books a viewing slot, and always routes sensitive conversations to a person.

2 Key Metrics

Trigger	5 webhooks called directly by Retell during a live phone call
Voice Platform	Retell — the caller talks to an AI agent, which calls these tools mid-conversation
Data Store	Google Sheets — Listings, Viewings and Escalations tabs
Office Notifications	Gmail, sent on every new listing, booked viewing and escalation
New Listings	Saved as pending_review — never published automatically, staff approve first
Viewing Conflicts	A property/date/time slot already booked is rejected before it reaches the sheet
Escalation Path	Financing, negotiation or anything out of scope routes straight to a human

5

VOICE TOOLS

3

SHEET TABS

0

AUTO-PUBLISHED LISTINGS

3 Business Value

- ✓ **Round-the-clock lead capture** — buyers get instant answers on availability any time they call, not just office hours.
- ✓ **New inventory funnel** — owners can list a property over the phone; it's queued for staff approval, not published blind.
- ✓ **No double-booked viewings** — every visit request is checked against existing bookings for that exact slot first.
- ✓ **Nothing sensitive handled by AI** — financing and negotiation are escalated straight to a human with full context.

4 Architecture Overview

Retell's voice agent is configured with five tools, each pointing at its own n8n webhook.

TOOL	CALLED WHEN	PURPOSE
check_property_availability	Caller asks what's available	Searches Listings by location, type, bedrooms, budget and sale/rent
list_property	Owner wants to list a property	Saves the property as pending_review and emails the office for approval
book_property_visit	Caller wants to view a property	Checks the requested slot is free, logs the viewing, emails the agent
get_property_details	Caller asks about one listing	Looks up a listing by ID or address and reads back price, size, description
request_human_callback	Anything out of the agent's scope	Logs an escalation and alerts the office with a one-tap call link

5 How It Works

Check Availability

- 1 Location, type, bedroom count, budget and buy/rent are pulled from the caller's request.
- 2 Listings already sold, rented or pending review are excluded automatically.
- 3 Location matching is partial and casual phrasing (buy/purchase) is normalized.
- 4 Up to 5 matches are summarized into a sentence Retell can speak directly to the caller.

List Property

- 1 Owner name, phone, property type, location, size, price, bedrooms and description are collected.
- 2 A unique property ID is generated and the listing is saved with status pending_review.
- 3 A formatted email goes to the office with every submitted detail for review.
- 4 The caller is told their property was submitted under their reference number.

Book a Visit

- 1 Customer name, phone, property ID and the requested date/time are captured.
- 2 Existing viewings for that exact property are checked for a conflict.
- 3 If free, the viewing is logged and the assigned agent is emailed the details.
- 4 If taken, no booking is made and the caller is offered another time.

Property Details

- 1 The caller's property ID (preferred) or a location/address is captured.
- 2 The Listings sheet is matched by exact ID, falling back to partial location.
- 3 Bedrooms, type, location, size, price and description are read back in one sentence.

Human Callback

- 1 Customer name, phone number and the escalation reason are recorded.
- 2 The request is logged to a dedicated Escalations tab with an urgent-styled office email.
- 3 The caller is told the request was flagged — this tool never transfers the live call.

6 Reliability & Security Design

- ◆ **No auto-published listings** — every submitted property lands as pending_review until staff approve it.
- ◆ **Server-side conflict checking** — viewing availability is verified against the sheet at booking time.
- ◆ **Tolerant matching, safe defaults** — natural phrasing is handled without over-matching sold or pending listings.
- ◆ **Clear escalation path** — financing, negotiation and anything out of scope guarantees a human follow-up.
- ◆ **Resilient notifications** — office and agent emails continue the workflow even if the send itself fails.
- ◆ **Full audit trail** — every listing, viewing and escalation is logged to Sheets and emailed immediately.

7 Setup Checklist

- ✓ Connect a Google Sheets credential with Listings, Viewings and Escalations tabs.
- ✓ Connect a Gmail credential and set the office/agent address on the three “Send” nodes.
- ✓ Activate the workflow and copy each webhook’s Production URL into the matching Retell tool.
- ✓ Pre-populate the Listings sheet with sample properties so search and lookups have real inventory.
- ✓ Test all five tools end-to-end with a live or simulated call.

8 Node-by-Node Map

check_availability	Check Availability Webhook → Normalize Criteria → Read Listings → Filter Matching Properties → Respond
list_property	List Property Webhook → Normalize Data → Append to Sheet → Send Office Email → Respond
book_property_visit	Book Visit Webhook → Check Slot → Append or Reject → Send Agent Email → Respond
get_property_details	Details Webhook → Read Listings → Find Property → Respond
request_human_callback	Escalation Webhook → Append to Sheet → Send Office Email → Respond

SUMMARY

Designed and built end-to-end in n8n, demonstrating a voice-agent backend for real estate with inventory search, slot-conflict checking, approval-gated listings, and human escalation.